

DNS — Guided Notes ANSWER KEY

Unit 1.6.2 | CompTIA Network+ N10-009 Obj. 1.6 | N10-008 1.6

For instructor use / distribute after students complete the notes.

Question 1

Match each DNS record type to its purpose:

A → _____

AAAA → _____

CNAME → _____

MX → _____

PTR → _____

Answer: A → maps hostname to IPv4 address; AAAA → maps hostname to IPv6 address; CNAME → alias pointing one hostname to another; MX → identifies mail servers for the domain; PTR → reverse lookup, maps IP address to hostname.

Question 2

A TXT record stores _____ data in DNS. Common uses include _____ (proves domain ownership to email providers) and _____ (specifies which servers are authorized to send email for the domain).

Answer: Arbitrary text; DKIM/DMARC verification; SPF (Sender Policy Framework).

Question 3

A forward lookup zone resolves _____ to _____. A reverse lookup zone resolves _____ to _____. Reverse lookups are used primarily for _____ and _____.

Answer: Hostnames to IP addresses; IP addresses to hostnames; logging/auditing and email spam verification (PTR checks).

Question 4

An authoritative DNS server holds the _____ records for a zone and gives definitive answers. A recursive resolver queries _____ on behalf of clients, caching results for up to the record's _____ value.

Answer: Actual/original; other DNS servers; TTL (Time to Live).

Question 5

DNS cache poisoning occurs when _____. An attacker who succeeds can redirect users to _____. The defense against this attack is _____, which adds cryptographic signatures to DNS responses.

Answer: An attacker inserts false DNS records into a resolver's cache; a malicious server (phishing, malware distribution); DNSSEC (DNS Security Extensions).

Question 6

Split-horizon DNS serves _____ DNS responses to internal clients and _____ DNS responses to external clients for the same domain name. This allows internal users to reach _____ while external

users reach _____.

Answer: Different; different; internal servers by private IP; the same service via public IP (or a DMZ address).

Question 7

The command _____ queries DNS and shows the record type, answer, and which server responded. The command _____ is a more detailed alternative available on Linux/Mac. To test a specific DNS server directly, the syntax is _____.

Answer: nslookup; dig; nslookup hostname server-ip (or dig @server-ip hostname).

Question 8 — Real World Application

REAL WORLD: A user can reach internal file servers by IP but not by name. External websites resolve fine. nslookup on their workstation returns correct results for external domains but times out for internal hostnames. What is the most likely cause and how do you fix it?

Answer: The workstation is querying an external DNS server (e.g., 8.8.8.8) instead of the internal DNS server, which hosts the internal zone. The external server has no records for internal hostnames. Fix: check the workstation's DNS server setting (ipconfig /all) — it should point to the internal DNS server. This is often caused by a DHCP misconfiguration (wrong DNS option) or a manual static DNS setting on the workstation. Correct the DNS server assignment and test with nslookup.